Senior AI Engineer –AKS-Accounting – Karbon – 29/11/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Sr. AI Engineer - Applied AI and LLM Solutions /Software Engineer/Full Stack Developer(Java/.NET/Python) /EDW /
Sr. DevOps Engineer /Sr. Data Engineer/Mulesoft/AI/ML/DL/BI/SP/D365 Developer - Japan (7.5yrs) and Australia (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior AI Engineer-Applied AI & LLM Solutions/Software Engineer/Sr. DevOps/Data Engineer
Responsibilities:

1. AI/ML Development & LLM Expertise

- Productionized complex Graph RAG and Agentic RAG pipelines that integrate advanced semantic search and vector technologies (e.g., Pinecone/Weaviate) to enhance multi-hop reasoning over complex documentation, achieving significant improvements in semantic query accuracy. (E.g.: graph_query_result = graph_rag.query('What is the environmental compliance history of facility A & its nearest flood zone?'))
- Drove the strategic planning and governance of AI initiatives by designing and overseeing Retrieval Augmented Generation (RAG) frameworks, ensuring large-scale knowledge bases are securely leveraged by LLMs while adhering to data governance and ethical standards.
- Strong proficiency in LLM development (for core reasoning and content generation) and specialized experience building autonomous agents using advanced orchestration frameworks like LangGraph, the OpenAI Agents SDK, and the Google ADK, ensuring complex, multi-step AI solutions are robust and production-ready.
- Architected and productionized bespoke AI/ML solutions (beyond standard LLM implementations) to streamline complex accounting payment workflows—generating actionable user insights and driving significant operational efficiencies. (E.g.: results = model.predict(payment_data_frame).to_json(orient='records'))
- Led the strategic design and development of AI solutions, consistently translating complex user problems into well-defined machine learning applications across traditional (GBM, regression) and advanced models.

2. Strategic Leadership, Architecture, and MLOps

- Senior AI/ML Technical Lead with 7+ years of end-to-end experience in the development, deployment, and optimization of production-grade AI solutions.
- Managed end-to-end solution ownership from concept through deployment, including defining and monitoring Key Performance Indicators (KPIs), while actively upskilling and educating junior engineers and client teams.
- Drove technical strategy by owning architectural decision-making and authoring detailed technical designs, while actively mentoring and coaching junior engineers in MLOps best practices and scalable solution development.
- Utilized exceptional problem-solving and communication skills to translate complex AI concepts into clear, actionable business language, successfully engaging stakeholders and driving technology adoption across both technical and non-technical client teams.
- Drove the successful productionization of AI solutions by owning the full MLOps cycle post-deployment, including comprehensive production monitoring. Collaborated closely with Data Engineers and Full Stack Developers to architect, build, and optimize the data and feature pipelines necessary for system reliability and performance.
- Elevated the Model Evaluation and Selection process by integrating criteria that account for business value, system stability, and long-term user impact, moving beyond basic accuracy metrics.

3. Backend Integration, Data Pipelines, and Cloud Infrastructure

- Designed and deployed highly scalable data transformation (ELT) pipelines, leveraging DBT for version-controlled, idempotent data modeling within Snowflake's Cloud Data Warehouse. Orchestration and CI/CD are managed through Azure Pipelines, delivering consumption-ready, high-quality datasets to C# backend services for powering production Machine Learning applications. (e.g., var features = await _featureStore.GetFeaturesAsync(userId); followed by var result = _modelClient.Predict(features);)
- Engineered and deployed optimized Python applications across a multi-cloud architecture, utilizing custom orchestration to automate, monitor, and accelerate critical data pipeline execution, improving deployment efficiency by 15%. (E.g.: await orchestrationService.execute_workflow(name='Data_Prep_Flow', input={'code': latest_commit.hash}))
- Expert in C# backend software development for building robust, high-performance integration layers and APIs required to deploy and serve complex ML/AI models in low-latency production environments. (e.g., [HttpPost("predict")] public async Task<ActionResult<Prediction>> Predict([FromBody] UserRequest request))
- Strong familiarity with Microsoft Azure cloud services, applying core infrastructure and platform features (e.g., Azure Kubernetes Service, App Services) to host, scale, and monitor end-to-end MLOps workflows and distributed backend AI applications. (e.g., az aks scale --name myCluster --node-count 5)
- Pioneered the use of Large Language Models (LLMs) and agentic workflows as optimizers within CI/CD pipelines to dynamically generate, test, and refactor code, resulting in a 25% reduction in deployment time. E.g.: llm_output = bedrock_client.invoke_model(modelId='anthropic.claude-4-sonnet-v1', body={"prompt": "Refactor this code to optimize performance: {code_snippet}"})
- Engineered and deployed scalable AI infrastructure using spec-driven agentic capabilities (AWS Kiro) and Infrastructure as Code (CDK/SAM/Terraform), automating MLOps pipelines for complex Graph RAG and vector database hosting while

building developer enablement tools (CLIs/templates) using GitOps, e.g., enabling cluster governance with kubernetes_version = "1.28" inside an EKS self-service template.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov Central Bank: | Software Engineer / Data Science & DevOps Specialist/ AI/ML Developer
Responsibilities:

1. High-Stakes Production AI and Multi-Agent Systems (MAS)

- AI-Driven Multi-Agent System (MAS): Architected and deployed a Python-based Multi-Agent System to autonomously monitor high-volume internal records (e.g., finance, user activity logs) in financial trading and settlement payment systems, successfully flagging high-risk anomalies indicative of account compromise and fraud. (E.g.: agent.dispatch_security_alert(risk_score=0.92))
- Production-Grade ML APIs: Engineered highly scalable, low-latency production-grade ML APIs in Python (Django, Flask) and Java (Spring Boot) to facilitate secure, real-time data exchange and continuous model deployment across the RBA's enterprise service layer, providing the groundwork for modern C# backend integration. (E.g.: POST /api/v1/model-inference HTTP/1.1 { "data": input_vector })
- Applied NLP and Computer Vision: Applied advanced techniques (NLTK, spaCy, OpenCV) for high-volume text analysis and non-textual anomaly detection within trading logs and multi-vendor settlement data flows, establishing early AI-driven security monitoring. (E.g.: nlp.analyze(text).anomalies.to_json())

2. Foundational AI Modeling and Architecture

- **Pioneering AI & Advanced Modeling Foundations**: Led the initial efforts in predictive modeling, integrating early deep learning concepts (e.g., initial TensorFlow/Keras for CNN/RNN) and classical ML pipelines to drive pattern discovery, time-series forecasting, and autonomous anomaly detection in sensitive financial data flows. (E.g.: model.fit(training_data, epochs=10, validation_split=0.2))
- **Architectural & Full-Stack Development**: Architected and managed the full-stack infrastructure for a Cyber Risk Management Platform (Node.js/React), consuming high-performance RESTful API endpoints to deliver real-time threat posture and dynamic Role-Based Access Control (RBAC) to the UI. (E.g.: axios.get('/api/risk/threat-posture'))
- Foundational RPA/Automation: Engineered intelligent rule-based automation (RPA) workflows powered by Python and NLP libraries (NLTK) to automate complex data analysis, preprocessing, and model training tasks at scale, streamlining core data support processes.

3. Early MLOps & Cloud Infrastructure

- **Pioneering MLOps & Multi-Cloud IaC**: Led the implementation of CI/CD pipelines for server-side ML application deployment, using scripting and configuration management (CloudFormation/boto3) to automate secure, scalable Infrastructure-as-Code (IaC) provisioning across initial multi-cloud environments (AWS, Azure, GCP). (E.g.: ansible-playbook deploy_low_latency_engine.yml)
- **Cloud-Native Architecture & Data Lakehouse**: Implemented cloud-native architectures utilizing AWS ECS for containerized microservices and built foundational components of a Data Lakehouse architecture (S3, Redshift) for efficient model serving and streamlined data ingestion, demonstrating vendor-agnostic cloud adaptability.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: | AI Pioneer / Data Engineer & Software Architect
Responsibilities:

1. Foundational AI, Predictive Modeling, and Anomaly Detection

- **Pioneering ML & Predictive Analytics**: Led the development of a predictive system for international student enrollment and finance flows (including auto currency changes for overseas payments). Engineered and deployed high-performance Machine Learning models using Python (Scikit-learn) and R within an early Cloud-based platform (Azure ML Studio), establishing core expertise in forecasting and model production. (E.g.: predicted_students = predict_enrollment(prospective_students.copy()))
- **Security & AI-Driven Anomaly Detection**: Developed predictive models and rule-based systems (precursors to MAS) in Python to autonomously monitor high-volume enrollment and financial records (multi-bank linked and auto-currency systems), flagging anomalies indicative of account compromise and fraud. (E.g.: agent.flag_anomaly(risk_profile))
- **Secure Enterprise APIs & Deep Learning Foundations**: Built secure enterprise APIs (Django/Flask) utilizing JWT for granular access control and applied diverse ML algorithms (ANN, regression) to secure and manage sensitive student data. Introduced foundational concepts related to Convolutional and Recurrent Networks (CNN/RNN) for pattern recognition.

2. Full-Stack Architecture and Cloud Enablement

- Enterprise Integration Architecture (Financial Systems): Served as an Integration Architect, designing and implementing a framework (MuleSoft/ESB) to seamlessly integrate the student enrollment and multi-bank payment application with disparate legacy systems (SAP, custom REST services) for high-volume data orchestration.
- **Full-Stack Development & High-Volume APIs**: Built scalable, secure backend services using Python (Flask) and early Node.js implementations. Focused on robust RESTful API development and data security protocols (helmet) for managing high volumes of concurrent applications.

- **DevOps, Automation, & Standardized Deployment**: Designed and maintained robust CI/CD pipelines using platform-specific tools, utilizing Python scripting to automate and streamline secure, standardized infrastructure deployment and customized ETL orchestrations. (E.g. (Cloud Sync/IaC precursor): aws s3 sync local_data/ s3://uc-student-data-bucket/ --delete)

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: **| Python ML Engineer & DevOps Automation Specialist /Software Test Engineer**
Key Achievements and Responsibilities:

- **Predictive ML Model Design & Development:**
- Designed and deployed Predictive Risk Models using Python and the Scikit-learn framework to accurately assess the likelihood of specific injury claim outcomes and potential fraud indicators.
- Developed and applied traditional ML techniques (e.g., implementing logic: high_risk_claims = claims[model.predict_proba(features)[:, 1] > 0.7]) to design predictive systems that automated workflows and generated critical claims insights.
- **ML Model Productionisation & Monitoring (MLOps):**
- Participated in the Machine Learning Development Lifecycle, deploying predictive models into the production environment, and utilizing Python and PowerShell scripting to automate system configuration and monitoring processes to ensure the stability of the claims analytics platform.
- Leveraged early cloud services (AWS EC2, S3, RDS) to build Foundational Infrastructure Automation, providing a scalable and high-performance environment for data and model services (precursor to modern MLOps and Production Monitoring).
- **Data Management & ETL/Data Pipelines:**
- Drove core data engineering initiatives by defining complex ETL (Extract, Transform, Load) requirements and creating robust data transformation logic to build the enterprise data model.
- Effectively handled heterogeneous data sources and formats (including structured Tabular data and complex location data) to provide high-quality, usable feature sets for downstream AI/ML models.
- **Core Application and API Development:**
- Developed core application features for the injury claim system using Python and RESTful APIs, e.g., implementing: app.post('/api/claims', async (req, res) => { // secure claim intake }), to optimize the user experience for claim management and dashboards.
- Ensured system quality and resilience by designing and executing comprehensive software test strategies, maintaining and debugging enterprise services using Python, Java, and PowerShell scripting for automation.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: | Data Engineer / Digital Transformation Lead
Responsibilities:

- **Data Pipelines and Enterprise Data Management:**
- Served as the key data specialist responsible for managing high-volume data integrity, including core Council Geospatial records (Vector/Tabular).
- Defined complex ETL requirements and data governance standards to build and maintain the enterprise data repository, implementing critical data validation logic such as: CASE WHEN Rate_Arrears_Amount IS NULL OR Rate_Arrears_Amount < 0 THEN 'Error: Arrears Check' ELSE 'Valid' END.
- Foundational Infrastructure Automation & Deployment:
- Managed system infrastructure (virtualized and early cloud instances like AWS EC2 and S3), utilizing Python scripting and PowerShell for configuration management and automated provisioning.
- Implemented early Continuous Integration (CI) processes (Jenkins) for automated builds, using scripts to enable faster deployment of data-driven applications.
- Software Quality & Backend Development:
- Led the software testing lifecycle and developed internal REST APIs using Python and Java to facilitate seamless data exchange across systems.
- Focused on robust logging, monitoring, and security measures, ensuring application reliability (e.g., performing early monitoring using AWS CloudWatch commands). Example: subprocess.run(["powershell", "-c", "Get-WmiObject Win32_Service | Where-Object {$_.State -ne 'Running'}"])

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo – Life Insurance Claim and Payment systems**
Finance/Insurance Domain: **DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DBA**
Responsibilities:

- **Foundational MLOps Platform & CI/CD for Policy Logic**:
- Spearheaded the implementation of early Continuous Integration (CI) and automated deployment processes, which established the CI/CD pipeline foundation necessary for rapid ML model retraining and deployment into production.

- **Infrastructure-as-Code (IaC) for AI/Data Scalability**:
- Utilized scripting and configuration management tools to implement Infrastructure as Code (IaC) principles, enhancing the scalability and reliability of the data warehouse (DWH) and core policy systems.
- Work directly supported timely risk analysis and claims processing, demonstrating foundational expertise in designing scalable infrastructure (AWS/Azure equivalent) required for modern high-volume data and ML workloads.
- **Automation for Risk & Cost Reduction:**
- Managed and automated critical DevOps and Unix/Linux systems supporting policy data, ensuring platform stability and high availability for core business and policy-related applications.
- Reduced operational risk and costs by deploying automation and self-service capabilities, minimizing infrastructure dependency and guaranteeing the continuity of service for data pipelines and customer systems.
- System Security and Data Governance:
- Administered and monitored core infrastructure, ensuring robust security measures were in place to safeguard sensitive customer and policy data.
- Maintained strict regulatory and data compliance standards, highlighting an understanding of data governance and security best practices essential for sensitive financial and accounting data.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo – Life Insurance Claim and Payment Systems**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer/ Unix/Linux Systems Administrator /EDW
Responsibilities:

- **Platform Modernization & Data Readiness**:
- Managed the end-to-end project to replace a legacy IBM AS/400-based application with a modern WISA-based system, a critical effort for enhancing the efficiency and compliance of new policy underwriting and customer data management.
- Led large-scale system refactoring.
- **Real-Time Data Flow for ML Features**:
- Successfully integrated the new insurance platform with the Salesforce Automation (SFA) platform, enabling real-time data flow for sales performance analysis.
- Provided underwriters with timely, accurate application data essential for risk assessment.
- **High-Availability Infrastructure for Data Warehousing (DWH)**:
- Planned, implemented, and administered core IT systems (Hyper-V, VMware, SAN Storage, Windows Server/Linux) that maintained the 24/7 high-availability of the Data Warehouse (DWH) and underwriting applications.
- **UNIX/Linux Administration & Security Foundation**: Administered and monitored critical UNIX/Linux systems, ensuring the smooth, secure operation of applications supporting financial data processing. Implemented security principles and core software management best practices. Example (Resource Check): df -h /var/log/ | awk '{print $5}'
  Example (Security Configuration): grep "LISTEN" /etc/ssh/sshd_config
- Ensured business continuity for policy processing and regulatory reporting, demonstrating foundational expertise in administering the high-performance infrastructure required for modern data and ML workloads.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo** - Business Intelligence (BI) & Analytics Systems
Software Vendor: Software Engineer /ML & Data Engineer /BI Developer /Data Warehouse Architect
Achievements and Responsibilities:

- **Pioneering ML Model Development & Deployment:**
- Designed, built, and deployed early machine learning models using k-nearest neighbors (k-NN) for critical business applications, specifically focusing on customer churn prediction and fraud detection.
- Achieved a demonstrable 15% improvement in prediction accuracy over previous methods.
- **Data Engineering for Enterprise ML/BI:**
- Led the planning, design, and development of large-scale enterprise data solutions, serving as the Data Warehouse Architect.
- Designed the data warehouse architecture, including schema definitions and complex Fact and Relational Dimensions tables, creating the scalable data foundation required for analytics and ML workloads.
- **Complex ETL Pipeline Development & Automation:**
- Engineered and automated complex ETL processes using SSIS (with T-SQL, C++, C#, VB.NET programming), transforming heterogeneous data and loading it into the DWH to produce Analysis Services Cubes (OLAP).
- Utilized SQL queries for advanced data extraction and manipulation, ensuring data accuracy and integrity across managed company databases.
- **Software Testing and Automation**:
- Ensured system quality through UAT and TEST cycles in QA environments, creating automated processes using VBScript and batch files to streamline the deployment and validation of data solutions.
- Demonstrated a commitment to quality and automation, which forms a vital part of the ML development lifecycle and MLOps practices.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos 8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0
- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI (ChatGPT-4o, Multi-GPTs, Auto-Agents), Amazon Bedrock (Claude/Titan) & SageMaker, Azure OpenAI Service, Azure ML, Azure AI Studio, Copilot, Azure AI Foundry, Google Vertex AI, Gemini API (formerly MakerSuite) , Agent-2-Agent
- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, Dynamics CRM 3.0~Dynamics 365 CE/D365 FO, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
- **Workflow & Automation Platforms**: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve

Senior AI / Machine Learning Engineer – Leading Finance Organization – TRC-Anne-Maria – 18/11/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Sr. AI/ML Engineer– Bedrock-SageMaker Specialist / GenAI | AI Build | Automation Engineer/ Sr. DevOps /Data Engineer
Software Engineer/Full Stack Developer(Python/Java/.Net/Mulesoft /AI/ML/DL/BI/SP/D365)-Japan (7.5yrs) & Aust. (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Sr. AI/ML Engineer–Bedrock-SageMaker Specialist / GenAI | Software Engineer | Automation Engineer
Responsibilities:

- **Professional Summary & Generative AI Strategy**
- Senior AI Engineer with 5+ years of experience in ML/Software Engineering and a 3-year foundation in Japanese Life Insurance operations and policy systems. Includes 2+ years dedicated to architecting and deploying production-grade Generative AI and LLM solutions. Proven hands-on expertise in Python, specializing in advanced RAG (including Graph RAG), prompt engineering, vector search, and model fine-tuning to drive business transformation across core insurance processes.
- Directed the end-to-end design, development, and high-availability deployment of production-grade Generative AI and LLM systems to build AI systems that redefine how customers engage across digital channels, automate key business use cases, and drive operational transformation.

- **Agentic AI & Advanced RAG Engineering**
- Engineered and productionized complex RAG and Agentic RAG pipelines, utilizing frameworks like LangGraph and LlamaIndex, integrating vector search (e.g., Pinecone/Weaviate) and Graph RAG to enhance multi-hop reasoning and

achieve a 25% improvement in semantic query accuracy.
E.g.: text_splitter = RecursiveCharacterTextSplitter(chunk_size=1000, chunk_overlap=200).

- Developed and orchestrated sophisticated autonomous AI agents using frameworks like LangGraph, AutoGen, and Semantic Kernel, and leveraging inter-agent communication protocols and tools like n8n to integrate with external APIs, driving operational efficiency and intelligent decision-making.
  E.g., response = await kernel.invoke_async("email_plugin.summarize", text=email_body).
- Achieved end-to-end cyber risk autonomy by architecting a coordinated system of Agentic AI Agents that translate qualitative Customer Needs into quantified, financial risk exposure using a proprietary Cyber Risk Quantification (CRQ) Engine.
- **Real-Time CX Automation & Backend Systems**
- Engineered and deployed Generative and Agentic AI solutions focused on real-time NLP and voice-to-text integration (e.g., using OpenAI's streaming Whisper API) to significantly enhance digital CX automation while simultaneously improving operational efficiency through call deflection and intelligent routing within CRM and customer journey platforms, resulting in sub-second transcription latency: stream = openai.Audio.transcribe('whisper-streaming', audio_file).
- Designed & deployed scalable backend microservices using Node.js in TypeScript and Python frameworks (Django, Flask), implementing cloud-native architectures with AWS ECS/EKS to manage non-blocking I/O. Enabled real-time data processing & seamless integration with AI-driven analytics pipelines built on platforms like LangChain, Amazon SageMaker, and Bedrock.
  E.g.1: app.post('/stream-data', (req: Request, res: Response) => { req.pipe(new DataProcessorStream()).pipe(res); });
  E.g.2: async function runAIPipeline(data: { source: string; }): Promise<{ prediction: string; confidence: number }> { return { prediction: 'anomaly_detected', confidence: 0.92 }; }
- **MLOps, Cloud Infrastructure & Production Deployment**
- Applied a builder's mindset to take full technical ownership of the LLMOps lifecycle, leveraging cloud platforms—specifically AWS (Bedrock, SageMaker)—to productionize scalable GenAI systems end-to-end, while automating CI/CD pipelines for 25% reduction in deployment time. E.g., boto3.client('lambda').invoke(FunctionName='AutomatedRemediationFunction', ...)
- Pioneered the use of Large Language Models (LLMs) and agentic workflows as optimizers within CI/CD pipelines to dynamically generate, test, and refactor code, resulting in a 25% reduction in deployment time and a significant improvement in code quality. E.g., llm_output = bedrock_client.invoke_model(modelId=' anthropic.claude-4-sonnet-v1', or 'amazon.titan-text-express-v1' body={ "prompt": "Refactor this code to optimize performance: {code_snippet}"
- Architected and deployed enterprise-grade Generative AI solutions on Amazon Bedrock, leveraging foundation models and integrating with core business systems via data-intensive APIs, reducing response time by 30%.
- Leveraged AWS Kiro's spec-driven development to design, build, and maintain scalable, secure AWS infrastructure (EC2, EKS, S3, RDS, Lambda, SageMaker, Amazon Neptune) for hosting advanced Graph RAG pipelines and LLMs.
- Developed, trained, and deployed a suite of machine learning and deep learning models on Amazon SageMaker using PyTorch, TensorFlow, and Hugging Face Transformers, automating model retraining to improve predictive accuracy and deliver NLP solutions.
- **Strategic Leadership & Outcome Delivery**
- Collaborated cross-functionally with Product, Engineering, and Data teams to ensure AI efforts delivered tangible business outcomes, always prioritizing system maintainability, performance, and responsible AI use.Strategically framed complex Aged Care insurance business challenges—including claims automation and policy optimization—into defined ML tasks, emphasizing model interpretability and regulatory compliance in the final architecture selection.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Sr. AI/ML Engineer/Automation Engineer/Sr. DevOps/Data Engineer /AI/ML/DL/Mulesoft Developer
Responsibilities:

- **Agentic AI & Financial Production Systems**
- Pioneered and deployed Agentic AI (Multi-Agent Systems), a subfield of distributed AI in Python, to automate and optimize complex algorithmic strategies for dynamic financial market pattern analysis and risk mitigation.
- Engineered scalable, mission-critical production applications in Python (leveraging Django/Flask) and robust system integrations, driving continuous deployment and MLOps for real-time machine learning services.
  E.g., requests.post(f"{api_url}/api/v1/deployments", json={'app':'ml-service', 'version':'v1.0.0'}).
- Built robust predictive models and secure APIs using a diverse Deep Learning algorithm suite (KNN, CNN, RNN, GAN, Scikit-learn, Keras) to drive risk mitigation and pattern discovery, directly enhancing algorithmic trading strategies.
- **Real-Time CX Automation & NLP**
- **Audio Intelligence**: Led end-to-end ML solution development, integrating speech-to-text APIs for high-volume call transcription, automated summarization, and sentiment analysis within CRM workflows, directly addressing Digital CX automation.
- Engineered Intelligent Automation workflows, powered by core NLP libraries (NLTK, spaCy), to automate lead qualification and critical customer interaction analysis, enhancing data-driven decision-making.
  E.g., ml_model.predict(vectorizer.transform(["RBA rate change triggers volatility"]))
- Developed Computer Vision models using OpenCV for automated validation of property documents and anomaly detection, streamlining regulatory compliance workflows.

- **MLOps, Cloud-Native Deployment & Low-Latency Backend**
- MLOps & CI/CD: Engineered and productionized mission-critical applications in Python (Django, Flask), establishing scalable IaC CI/CD pipelines and isolated ML platforms to ensure robust deployment across AWS, Azure, and GCP.
- **Cloud Architecture & Microservices**: Implemented a cloud-native architecture using AWS ECS for containerized microservices and REST APIs, enabling efficient model integration with enterprise platforms and streamlined data ingestion.
- **Low-Latency Automation**: Specialized in designing and implementing end-to-end automation solutions using Python and Ansible to streamline secure, low-latency infrastructure provisioning for RBA trading platforms.
  E.g., ansible-playbook deploy_low_latency_trading_engine.yml to automate trading engine deployments.
- API Gateway Expertise: Developed and deployed scalable applications using MuleSoft Anypoint Platform, focusing on robust backend API development and seamless system integrations for core banking systems.
  E.g. 1: POST /api/v1/orders HTTP/1.1 { "item": "widget", "qty": 2 }

Oct. 2013 – Nov.2015 University of Canberra (UC), ACT(Remote from SYD) - - Advanced ML/AI Systems and Automation
Gov. University: Sr. ML Engineer/Automation Engineer /Integration Solution Architect /ML/DL/Mulesoft Developer
Responsibilities:

- **Automated Systems and Machine Learning**
- Led the design and implementation of highly available, automated Machine Learning and AI solutions to optimize core university operations, focusing on system performance and data-driven insights.
- **Core ML Engineering & Applied AI**
- Designed and engineered highly available Multi-agent Systems (MAS) in Python for the autonomous management and adaptation of complex data flows, demonstrating expertise in advanced computational models & self-learning systems.
- Developed intelligent rule-based Robotic Process Automation (RPA) workflows, utilizing Natural Language Processing (NLP) libraries (e.g., NLTK, spaCy) for automated data classification, analysis, and interaction processing.
- Led end-to-end Machine Learning project development, encompassing comprehensive data collection, automated transcription, statistical analysis, and intelligent summarization, ensuring high data integrity and efficient pipelines for research and academic projects.
- Developed and optimized Computer Vision models (using OpenCV and Scikit-image) for integrating visual data analysis and sophisticated anomaly detection into existing automation pipelines.
- **Production-Grade Model Development & Scaling**
- Built robust, predictive models and secured APIs using a diverse range of machine learning algorithms (KNN, CNN, RNN, ANN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning).
- Leveraged core ML frameworks like scikit-learn, NLTK, Theano, and Caffe to develop scalable Python applications for data preprocessing, modeling, and large-scale natural language processing tasks.
- Optimized machine learning model training and inference by utilizing GPU-accelerated systems, significantly improving computational efficiency for large-scale research projects.
- Applied emerging principles in statistical analysis to develop and rigorously test algorithmic strategies, ensuring model validity and predictive accuracy across various systems.

- **Systems Integration & Cloud Architecture**
- Implemented scalable cloud-based architectures using AWS ECS for deploying containerized microservices and REST APIs, ensuring seamless model integration and high-throughput data ingestion pipelines.
- Integrated third-party speech-to-text services for automatic transcription and sentiment analysis, developing custom-built data analysis pipelines to contextualize unstructured data.
- Managed and analyzed large datasets by integrating and utilizing databases such as PostgreSQL and MongoDB, supporting real-time data processing and context-aware insights for organizational decision-making.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: ML Engineer /Automation Engineer/Software Test Engineer /DevOps/Data Engineer
Responsibilities:

- Specialized in building Machine Learning (ML)-driven automation and predictive analytics platforms, directly optimizing the financial claims lifecycle for a high-volume compensation insurer, delivering significant gains in risk management and processing efficiency.
- **Intelligent Automation & Multi-Agent System (MAS) Foundations for Financial Automation**
- Claims Automation, Risk Modeling & Efficiency
- Pioneered Claims Processing Automation: Developed self-learning Multi-Agent Systems (MAS) in Python to autonomously optimize and automate key stages of the injury claims processing lifecycle, significantly improving overall system efficiency and reducing manual claim handling time.

- **Predictive Claims Risk Modeling**: Developed and deployed predictive models (scikit-learn) to accurately assess the likelihood of specific claim outcomes, claim duration, and potential fraud indicators.
- **Business Impact**: Enabled proactive risk management and optimized resource allocation across the entire injury compensation claims domain.
- **Scalable Claims Data Management**: Engineered secure, high-performance backend services using Python (Flask) and Java to efficiently manage the vast volume of injury compensation claim data, ensuring high data integrity and seamless integration with critical claims management systems.
- Optimized Claimant Experience: Designed and implemented user interfaces (jQuery, JavaScript) that optimized the claimant experience for EML's online portal and streamlined data presentation for complex claims workflows, reducing ambiguity for adjusters.
- **Data Architecture & Claims Analytics**
- Real-Time Claims Analytics Platform: Architected and implemented data solutions on AWS (EC2, RDS, S3), focusing on traditional data warehousing (Redshift) and real-time ETL processes (Kinesis) to support advanced claims analytics and regulatory reporting.
- **Big Data Processing**: Utilized Amazon EMR for batch processing of large data volumes, enabling sophisticated querying of the claims data lake (Redshifttrum) for advanced fraud and exposure reporting.
- Data Governance & Mapping: Created and implemented detailed mapping documents defining data transformation, loading, and model requirements, ensuring strict data governance for complex claims data fields.
- **DevOps & CI/CD for Claims Systems**
- Automated Deployment of ML Solutions: Championed Continuous Integration and Delivery (CI/CD) pipelines (Jenkins) to automate the secure deployment of scalable ML applications and services, ensuring high availability of claims models and infrastructure (AWS EC2).
- **Operational Consistency**: Utilized PowerShell scripting to automate system configurations, testing, and deployment processes, adhering to rigorous software development best practices for versioning and quality control across claims-critical applications.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia -** FinTech Infrastructure & MLOps Foundation
Government Council: DevOps/Data Engineer /Data Engineer /Software Test Engineer /System Integration Specialist
Responsibilities:

- Built the scalable software and data infrastructure for high-volume government financial transaction systems, establishing early DevOps/CI/CD practices essential for the eventual deployment of predictive and automation models.
- **Foundational MLOps & Production Deployment**
- Pioneered MLOps Environment: Designed and implemented automated CI/CD pipelines (Jenkins, shell scripting) to manage server configurations (Puppet) and ensure the reliable, high-availability delivery of both web applications and early-stage machine learning models for research projects.
- Engineered Production-Ready APIs: Architected and developed robust REST APIs using Python and Java to manage and integrate the core business logic for high-volume financial payment systems, establishing the microservice foundation required for modern AI/ML integration.
- Scalable Infrastructure Management: Deployed and maintained web applications and data services on virtualized servers, utilizing scripting languages and automation tools to manage infrastructure configurations and ensure system stability.

- **Data Integrity & Governance for Financial Systems**
- **Data Management & Integration**: Managed high-volume financial transaction data using SQL and relational databases, ensuring data integrity, optimized retrieval, and strict governance necessary for audit and analytical reporting.
- **System Integration**: Implemented and managed APIs to ensure seamless, compliant integration with diverse external government and financial systems, maintaining consistency across complex data landscapes.
- **Security & Compliance**: Implemented security best practices, including regular system and application patching, and maintained logging and monitoring processes to ensure the security and compliance of public-facing financial applications.

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Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo -** Core FinTech Infrastructure & MLOps Foundation
Finance/Insurance Domain: DevOps Engineer/Sr. Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Drove the modernization of core life insurance financial systems through DevOps and automation, establishing the production-ready infrastructure necessary for scalable data processing and the rapid deployment of future risk and pricing models.
- **MLOps Foundation & Continuous Deployment**
- **Accelerated Financial Feature Deployment (CI/CD)**: Spearheaded the implementation of Continuous Integration (CI) and automated deployment processes, significantly reducing the delivery time for new software features and regulatory updates within policy administration systems.

- **Direct ML Relevance**: This established the CI/CD pipeline foundation critical for rapid, secure, and repeatable Machine Learning (ML) model deployment (CI/CD for MLOps).
- **Infrastructure for Scalable Risk Modeling (IaC)**: Utilized advanced scripting and configuration management tools to implement Infrastructure as Code (IaC) principles, enhancing the scalability and reliability of the Data Warehouse (DWH) and core policy systems, which are crucial for timely financial risk analysis and claims processing.
- **Reduced Operational Risk via Automation**: Maintained the stability and high availability of critical Unix/Linux systems supporting sensitive financial and policy data, applying robust automation to ensure business continuity and reduce operational risk.
- **High-Availability & Regulatory Compliance**
- **System Security for Compliance**: Established and enforced robust security measures across expanding infrastructure to safeguard sensitive customer and financial data, ensuring strict adherence to regulatory and data compliance standards expected in the finance sector.
- **Business Continuity and Support**: Administered and monitored core systems and provided high-level technical support, guaranteeing the smooth, continuous operation of all critical business and policy-related applications handling complex financial transactions.
- **Technology Environment**: Deep expertise in Unix/Linux system administration, enterprise-level Microsoft infrastructure (AD, Exchange), and Data Warehousing (DWH) management, providing a strong technical backbone for complex financial data pipelines.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo -** FinTech Modernization & Data Flow Integration
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator
Responsibilities:

- Managed the modernization of core financial systems, establishing a scalable, data-integrated platform that improved the speed of underwriting decisions and built the high-availability infrastructure required for future risk modeling and predictive analytics.
- **Platform Modernization for Data & Risk Assessment**
- **Underwriting System Modernization (ML Data Foundation)**: Led the critical project to replace the legacy system with a modern platform, establishing a scalable data environment essential for the ingestion and processing of data needed to train and deploy advanced ML models for fraud/risk assessment.
- **Real-Time Data Flow for Risk Analysis**: Successfully managed the integration of the new insurance platform with the Salesforce Automation (SFA) platform, creating a real-time data flow that provided timely, accurate application data essential for risk assessmen.
- **Project Leadership & Compliance**: Applied strong project management and systems engineering skills to ensure the complex migration was completed on time, maintaining strict compliance and customer data management standards throughout the process.
- **High-Availability Infrastructure & Business Continuity**
- **Infrastructure for High-Availability MLOps**: Planned, implemented, and administered core, high-availability infrastructure (Hyper-V, SAN, Windows/Linux) that ensured 24/7 uptime for the Data Warehouse (DWH) and underwriting systems, demonstrating expertise in managing the production environment necessary for scalable ML models and REST APIs in finance.
- **Business Continuity for Policy Processing**: Ensured business continuity for critical policy processing and regulatory reporting systems, demonstrating operational stability required in high-stakes financial operations.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo** - Early ML Modeling & Enterprise Data Systems
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Designed and developed large-scale Business Intelligence solutions, establishing enterprise data architectures and applying Machine Learning to deliver early predictive analytics and risk mitigation capabilities for customers.
- **Predictive Modeling & Risk Analytics**
- Pioneered Early Predictive ML: Built and deployed machine learning models (specifically k-Nearest Neighbors - k-NN) for critical financial applications focused on customer churn prediction and fraud detection.
- **Quantifiable Impact**: Achieved a 15% improvement in prediction accuracy for these models, directly contributing to enhanced risk management and customer retention strategies.
- **Production Deployment Foundation**: Involved in the UAT and TEST cycles, using VBScript and batch files to automate QA processes and validate the quality of the predictive solutions before production release.
- **Enterprise Data Engineering & System Foundation**
- **Scalable Data Architecture for M**L: Designed and built Data Warehouse (DWH) architecture (including hardware and schema definition) using MS BI tools (SSIS, SSAS, SSRS), creating the necessary clean, structured data environment for large-scale predictive modeling.

- **Complex ETL for Feature Engineering**: Utilized SSIS ETL tools to extract and transform complex data, effectively performing the feature engineering required to train and populate the Analysis Services Cubes and Data Marts that fed the predictive models.
- **Software Development for Production**: Developed enterprise-level packaged software using C++, C#, VB.NET, and T-SQL, demonstrating expertise in building, debugging, and maintaining production-grade software systems.
- Database Management: Developed complex SQL queries for data extraction and transformation, and managed company databases, ensuring the data accuracy and integrity essential for reliable model performance.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos 8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0
- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI (ChatGPT-4o, Multi-GPTs, Auto-Agents, OpenAI API), Amazon Bedrock (Claude/Titan) & SageMaker, Azure OpenAI Service, Azure ML, Azure AI Studio, Copilot, Azure AI Foundry, Google Vertex AI, Gemini API (formerly MakerSuite) , Agent-2-Agent
- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, CRM 3.0~D365, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
- **Workflow & Automation Platforms**: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve

AI Engineer - Applied AI and LLM Solutions –AKS,OpenAI- CorrelateRecruitment- Kiara 16/11/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
AI Engineer - Applied AI and LLM Solutions /Software Engineer/Full Stack Developer(Java/.NET/Python) /EDW /
Sr. DevOps Engineer /Sr. Data Engineer/Mulesoft/AI/ML/DL/BI/SP/D365 Developer - Japan (7.5yrs) and Australia (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: AI Engineer - Applied AI and LLM Solutions /Software Engineer/Snr DevOps/Data Engineer/ AI/ML Developer
Responsibilities:

1. Strategic Consulting, Leadership & AI Product Ownership

- Leveraged 5+ years of strategic consulting and solution design experience to function as the critical bridge between executive business objectives and complex, production-ready AI engineering delivery.
- Managed end-to-end solution ownership from concept through deployment, including defining and monitoring Key Performance Indicators (KPIs), while actively upskilling and educating client organizations on new Microsoft Fabric/Databricks AI capabilities via bespoke workshops and training sessions.
- Spearheaded the development and execution of comprehensive digital transformation roadmaps, successfully integrating AI, automation, and Azure-based digital platforms to achieve key business goals such as reducing operational costs and accelerating time-to-market.
- Utilized exceptional problem-solving and communication skills to translate complex AI concepts into clear, actionable business language, successfully engaging stakeholders and driving technology adoption across both technical and non-technical client teams.

2. AI Architecture, LLM Solutions & Responsible AI

- Drove the strategic planning and governance of AI initiatives by designing and overseeing Retrieval Augmented Generation (RAG) frameworks, ensuring large-scale knowledge bases are securely leveraged by LLMs while strictly adhering to NSW AI and data governance frameworks and ethical standards.
- Architected and delivered production-grade AI solutions leveraging a multi-cloud foundation (Azure, Databricks, AWS) and API-based deployments, pioneering LLM, RAG, and agentic workflows within MLOps pipelines.
- Partnered directly with business executives to understand pain points, designing and implementing Human-in-the-Loop (HITL) systems and governance frameworks within Azure Data Services to ensure LLM outputs remained accurate and compliant, driving measurable process improvements.
- Productionized complex Graph RAG and Agentic RAG pipelines that integrate advanced semantic search and vector technologies (e.g., Pinecone/Weaviate) to enhance multi-hop reasoning over complex audit and geospatial documentation, achieving a 25% improvement in semantic query accuracy. E.g.: graph_query_result = graph_rag.query('What is the environmental compliance history of facility A & its nearest flood zone?')
- Spearheaded Intelligent Document Processing (IDP) projects by integrating Optical Character Recognition (OCR) and orchestrating RPA processes via API integrations to automate complex, rule-based tasks.
- Pioneered the use of Large Language Models (LLMs) and agentic workflows as optimizers within CI/CD pipelines to dynamically generate, test, and refactor code, resulting in a 25% reduction in deployment time.
  E.g.: llm_output = bedrock_client.invoke_model(modelId='anthropic.claude-4-sonnet-v1', body={"prompt": "Refactor this code to optimize performance: {code_snippet}"})

3. Data Architecture, MLOps Engineering & Geospatial Expertise

- Applied best practices in MLOps for streamlined model deployment, complemented by hands-on experience utilizing LangChain for GenAI orchestration and integrating with leading BI tools (e.g., Power BI) for measurable analytics delivery.
- Engineered and deployed optimized Python applications across a multi-cloud architecture, utilizing n8n orchestration to automate, monitor, and accelerate critical data pipeline execution, improving deployment efficiency by 15%.
  E.g.: await n8n.execute_workflow(name='CI_CD_Optimize', input={'code': latest_commit.hash})
- Architected and implemented production-grade Data Science solutions using Python, leveraging advanced libraries (e.g., Pandas, Scikit-learn, GeoPandas) to integrate complex spatial and non-spatial datasets, driving precision in analytical modeling. E.g.: spatial_df = non_spatial_df.merge(geopandas.read_file('audit_boundaries.geojson'), on='site_id', how='left')
- Expertly leveraged Topological and Raster Digital Elevation Models (DEMs) to architect efficient spatial databases, optimizing data structures for performance-critical GIS analysis and seamless integration into downstream analytical models.
  E.g.: SELECT ST_Slope(rast) FROM dem_imagery WHERE ST_Intersects(rast, asset_boundary) AND asset_id = 42
- Pioneered a hybrid data integration strategy for 1500 annual Aged Care Site Audits by seamlessly combining high-precision Vector data with rich Raster data, yielding a 20% increase in map accuracy and audit detail.
- Designed and operationalized high-throughput ETL pipelines using Python to seamlessly ingest, transform, and manage large-scale, heterogeneous data, ensuring robust data quality.
- E.g.: df_transformed = pd.DataFrame(rasterio.open(file).read(1).flatten()).merge(vector_data, on='id')
- Engineered and deployed scalable AI infrastructure using spec-driven agentic capabilities (AWS Kiro) and Infrastructure as Code (CDK/SAM/Terraform), automating MLOps pipelines for complex Graph RAG and vector database hosting while building developer enablement tools (CLIs/templates) using GitOps, e.g., enabling cluster governance with kubernetes_version = "1.28" inside an EKS self-service template.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov Central Bank: | Software Engineer / Data Science & DevOps Specialist/ AI/ML Developer
Responsibilities:

- **Pioneering AI & Advanced Modeling Foundations**: Led the initial efforts in predictive modeling, integrating early deep learning concepts (e.g., initial TensorFlow/Keras for CNN/RNN) and classical ML pipelines to drive pattern discovery, time-series forecasting, and autonomous anomaly detection in financial data flows.
  E.g.: model.fit(training_data, epochs=10, validation_split=0.2)
- **Applied NLP and Computer Vision**: Applied advanced techniques (NLTK, spaCy, OpenCV) for high-volume text analysis, intelligent summarization, and non-textual anomaly detection within trading logs and financial data flows, establishing early AI-driven security monitoring. E.g.: nlp.analyze(text).anomalies.to_json()
- **Production-Grade ML APIs**: Engineered scalable, low-latency production-grade ML APIs in Python (Django, Flask) and Java (Spring Boot) to facilitate secure, real-time data exchange and continuous model deployment across the RBA's enterprise service layer. E.g.: POST /api/v1/model-inference HTTP/1.1 { "data": input_vector }
- **AI-Driven Multi-Agent System (MAS)**: Architected and deployed a Python-based Multi-Agent System to autonomously monitor high-volume internal records (e.g., finance and user activity logs), flagging high-risk anomalies indicative of account compromise and fraud. E.g.: agent.dispatch_security_alert(risk_score=0.92)
- **Architectural & Full-Stack Development**: Architected and managed the full-stack infrastructure for a Cyber Risk Management Platform (Node.js/React), consuming RESTful API endpoints to deliver real-time threat posture (e.g., SQL Injection, Phishing) and dynamic Role-Based Access Control (RBAC) to the UI. E.g.: axios.get('/api/risk/threat-posture')
- **Foundational RPA/Automation**: Engineered intelligent rule-based automation (RPA) workflows powered by Python and NLP libraries (NLTK) to automate complex data analysis, preprocessing, and model training tasks at scale, streamlining core data support processes.
- **Pioneering MLOps & Multi-Cloud IaC**: Led the implementation of CI/CD pipelines for server-side ML application deployment, using scripting and configuration management (CloudFormation/boto3) to automate secure, scalable Infrastructure-as-Code (IaC) provisioning across initial multi-cloud environments (AWS, Azure, GCP).
  E.g.: ansible-playbook deploy_low_latency_engine.yml
- **Cloud-Native Architecture & Data Lakehouse**: Implemented cloud-native architectures utilizing AWS ECS for containerized microservices and built foundational components of a Data Lakehouse architecture (S3, Redshift) for efficient model serving and streamlined data ingestion.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: | AI Pioneer / Data Engineer & Software Architect
Responsibilities:

- **Pioneering ML & Predictive Analytics**: Led the development of a system to predict international student enrollment. Engineered and deployed high-performance Machine Learning models using Python (Scikit-learn) and R within an early Cloud-based platform (Azure ML Studio). E.g.: predicted_students = predict_enrollment(prospective_students.copy())
- **Security & AI-Driven Anomaly Detection**: Developed predictive models and rule-based systems (precursors to MAS) in Python to autonomously monitor high-volume enrollment and finance records, flagging anomalies indicative of account compromise and fraud. E.g.: agent.flag_anomaly(risk_profile)
- **Secure Enterprise APIs & Deep Learning Foundations**: Built secure enterprise APIs (Django/Flask) utilizing JWT for granular access control and applied diverse ML algorithms (ANN, regression) to secure and manage sensitive student data. Introduced foundational concepts related to Convolutional and Recurrent Networks (CNN/RNN) for pattern recognition.
- **Enterprise Integration Architecture**: Served as an Integration Architect, designing and implementing a framework (MuleSoft/ESB) to seamlessly integrate the student enrollment application with disparate legacy systems (SAP, custom REST services) for high-volume data orchestration.
- **DevOps, Automation, & Standardized Deployment**: Designed and maintained robust CI/CD pipelines using platform-specific tools, utilizing Python scripting to automate and streamline secure, standardized infrastructure deployment and customized ETL orchestrations.
  E.g. (Cloud Sync/IaC precursor): aws s3 sync local_data/ s3://uc-student-data-bucket/ --delete
- **Full-Stack Development & High-Volume APIs**: Built scalable, secure backend services using Python (Flask) and early Node.js implementations. Focused on robust RESTful API development and data security protocols (helmet) for managing high volumes of concurrent applications.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: | Python Application Developer /DevOps Engineer/Data Engineer /Software Test Engineer
Responsibilities:

- Predictive Risk Modeling & Applied AI: Developed and deployed advanced predictive models using Python (Scikit-learn) to accurately assess the likelihood of specific injury claim outcomes and fraud indicators. This foundational work informs current RAG/LLM applications in risk assessment.
  E.g. (Python Logic): high_risk_claims = claims[model.predict_proba(features)[:, 1] > 0.7]
- Core Application & API Development: Contributed to the development of the injury claim system, designing and implementing features using Python and full-stack JavaScript (e.g., AngularJS/jQuery) to streamline user experience for claim management and dashboard consumption via robust RESTful APIs.

- E.g. (API Development): app.post('/api/claims', async (req, res) => { /* secure claim intake */ });
- Data Engineering for Heterogeneous Data: Drove core data initiatives by defining complex ETL requirements (analyzing sources and formats) and creating robust transformation logic to build the enterprise data model, implicitly handling complex location and structured (Tabular) data critical for claims analysis.
- Test Engineering & Automation: Ensured system quality and resilience by designing and executing comprehensive software test strategies, maintaining and debugging enterprise services across languages including Python, Java, and PowerShell scripting for automation.
- Foundational Cloud & Infrastructure Automation: Designed and developed foundational data solutions using early cloud services (AWS EC2, S3, RDS) to ensure the scalability and performance of the claim analytics platform, utilizing PowerShell scripting and Python to automate the provisioning and configuration of production systems (precursor to modern IaC/DevOps).

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: | Data Engineer / Digital Transformation Lead
Responsibilities:

- Geospatial Data Engineering & Data Governance: Served as the key data specialist responsible for managing high-volume data integrity, including core Council Geospatial records (Vector/Tabular). Defined complex ETL requirements and governance for critical council data within the enterprise repository (SharePoint/Databases).
- E.g. (Data Integrity): SELECT column_name, data_type FROM information_schema.columns WHERE table_name = 'council_data'
- Software Quality & API Development: Led the software testing lifecycle and developed internal REST APIs using Python and Java to facilitate seamless data exchange across systems. Focused on robust logging, monitoring, and security measures to ensure application reliability. E.g. (API Development): app.route('/api/v1/status', methods=['GET'])
- Continuous Integration & Scripted Automation: Designed and implemented early Continuous Integration (CI) processes (Jenkins) for automated builds. Developed extensive Python and PowerShell scripts to automate system provisioning and configuration, enabling faster deployment of data-driven applications.
  E.g. (Automation): subprocess.run(["powershell", "-file", "deploy_app.ps1"])
- Foundational DevOps & Infrastructure Automation: Managed system infrastructure (virtualized and early cloud instances like AWS EC2 and S3), utilizing Python scripting and PowerShell for configuration management and automated provisioning—foundational work for later Infrastructure-as-Code (IaC) practices.
  E.g. (Monitoring): aws cloudwatch get-metric-statistics --namespace AWS/EC2

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- **Automation & Scripted Delivery (Precursor to CI/CD)**: Led the improvement of continuous integration and delivery processes through advanced Shell and Python scripting, successfully reducing software feature delivery time and ensuring deployment consistency across environments. Example: bash deploy.sh --target=production --service=financial-reporting
- **Data Warehouse (DWH) Administration**: Functioned as a DBA, managing and monitoring the core Data Warehouse infrastructure for critical financial and insurance data. Ensured high availability, integrity, and performance necessary for regulatory reporting and business intelligence. Example: psql -c "VACUUM FULL ANALYZE;"
- **Systems Resilience & Security**: Served as the Senior Systems Engineer for Unix/Linux environments, responsible for the 24/7 operation and security hardening of servers hosting mission-critical applications in the finance domain.
  Example: cat /etc/sysconfig/iptables | grep 8080
- **Operational Efficiency**: Automated Tier 1 and Tier 2 support functions by designing and deploying the VMM Self-Service Portal, which optimized resource allocation and significantly minimized IT Helpdesk running costs.
  Example (Virtualization Management): Get-VM | Start-VM -TargetHost "HyperVHost01"
- **Enterprise Infrastructure Support**: Provided Level 2/3 technical support and administration for core Microsoft enterprise infrastructure, including Active Directory (AD), DNS, DHCP, Exchange 2010, and SharePoint.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /EDW
Responsibilities:

- **Technology Transformation & Project Leadership**: Managed a critical initiative to replace legacy insurance applications (IBM AS/400 and LAMP stack) with a modern WISA-based platform. Strategic project was delivered to significantly reduce IT operational costs and enable integration with cloud-based Sales Force Automation (SFA). Example: SELECT platform_status, COUNT(*) FROM core_systems GROUP BY platform_status
- **Infrastructure Architecture & Systems Resilience**: Provided expert systems planning and implementation for all core IT infrastructure, including Hyper-V/VMware ESXi virtualization and high-availability SAN storage (HP/EMC/Dell).

Example (Virtualization Management): Get-VMHost | Select-Object Name, ProcessorType, VmCount

- **UNIX/Linux Administration & Security Foundation**: Administered and monitored critical UNIX/Linux systems, ensuring the smooth, secure operation of applications supporting financial data processing. Implemented security principles and core software management best practices. Example (Resource Check): df -h /var/log/ | awk '{print $5}'
  Example (Security Configuration): grep "LISTEN" /etc/ssh/sshd_config
- **Data Warehouse (DWH) Support**: Acted as a key operational resource for the Data Warehouse environment, overseeing the core infrastructure that supported the company's vital business and financial reporting systems.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH**
Responsibilities:

- **Large-Scale Data Warehouse (DWH) Architecture**: Led the planning, design, and development of complex, enterprise-level DWH solutions, including designing the hardware infrastructure landscape and defining detailed schemes, Fact and Relational Dimensions tables (Extended Star Schema), ensuring a robust architecture for global reporting requirements.
- **Enterprise ETL & Data Engineering**: Architected and built foundational, large-scale data pipelines (ETL using SSIS, T-SQL, C#, and VB.NET) essential for the organization's digital transformation towards advanced analytics. Pipelines processed complex raw data and loaded cleansed Master Data into Analysis Services Cubes (SSAS) for reporting.
- Established early enterprise standards for data quality assurance and cleansing, directly supporting today's requirements for governance, assurance, and reliable data lifecycle management (as required by current NSW AI and data policies).
- **Early Machine Learning (ML) Models**: Developed and deployed initial ML models using the k-nearest neighbors (k-NN) non-parametric supervised learning algorithm and Python for high-impact financial applications, including customer churn prediction and fraud detection.
- Business Intelligence (BI) & Reporting: Developed high-value BI solutions using the Microsoft stack (SSAS, SSRS). Managed reporting deployments and maintained underlying company databases, ensuring data accuracy and integrity for critical stakeholder requirements.
  Example (Querying Cubes): SELECT platform_status, COUNT(*) FROM core_systems GROUP BY platform_status

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos 8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0
- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI (ChatGPT-4o, Multi-GPTs, Auto-Agents), Amazon Bedrock (Claude/Titan) & SageMaker, Azure OpenAI Service, Azure ML, Azure AI Studio, Copilot, Azure AI Foundry, Google Vertex AI, Gemini API (formerly MakerSuite) , Agent-2-Agent
- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, Dynamics CRM 3.0~Dynamics 365 CE/D365 FO, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
- **Workflow & Automation Platforms**: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve